Juan David Muñoz-Bolaños

Application Engineer Image Processing

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PROFESSIONAL SUMMARY

Experienced Physics Engineer specialized in industrial machine vision, optical design, and hardware-software integration. Proven background in commissioning custom vision solutions on production lines. Expert in choosing the right illumination, cameras, and lenses to ensure high-performance quality assurance in automated manufacturing environments.

TECHNICAL COMPETENCIES

Machine Vision Hardware
Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

Image Processing Libraries
Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

System Alignment & Justage
Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

Automation & Integration
Skilled in automating optical systems with Python, C++, and LabVIEW. Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

R&D Methodology
Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

PROFESSIONAL EXPERIENCE

INTECOL S.A.S. — Medellín, Colombia
Junior Machine Vision System Engineer — *Mar 2018 – Mar 2019*

- **Vision Pipeline Engineering:** Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.
- **Industrial Integration:** Commissioned Cognex VisionPro systems for auto-part recognition at Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.
- **QA & Feasibility:** Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher – Optical Metrology & Automation — *Jan 2022 – Present*

- **System Build & Justage:** Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.
- **Adaptive Optics Control:** Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.
- **Vendor & Component QM:** Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R&D Researcher (Photonics & Data) — *Jun 2020 – Dec 2021*

- **Instrumentation Design:** Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.
- **Data Evaluation Hooks:** Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

EDUCATION

Medical University of Innsbruck — Austria
Ph.D. in Adaptive Optics & Microscopy — *Jan 2022 – Present*

Friedrich Schiller University Jena — Germany
M.Sc. in Photonics — *2019 – 2021*

University of Cauca — Colombia
B.Sc. in Physics Engineering — *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards:** ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte — Medical University of Innsbruck
Head of Institute
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Dr. Christoph Krafft — Leibniz Institute of Photonic Technology
Spectroscopy Group Leader
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